

UCI Adult Income Dataset Analysis Report

1. Data Cleaning and Preprocessing

The dataset was cleaned by removing duplicates and handling missing values. Categorical variables were encoded using Label Encoding, and numerical features were standardized.

2. Model Development and Evaluation

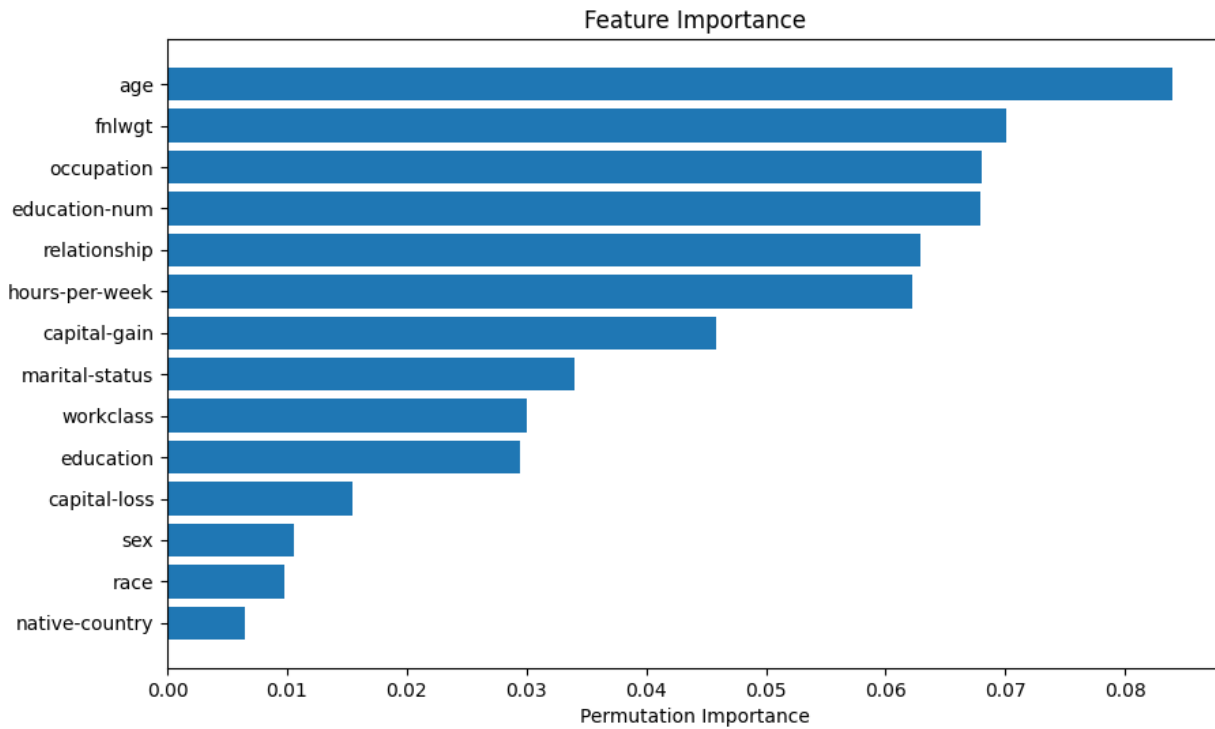
A Random Forest Classifier was used to predict whether an individual's income exceeds \$50K. The model was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC metrics. The model achieved the following performance metrics:

- Accuracy: 0.85
- Precision: 0.55
- Recall: 0.39
- F1 Score: 0.45
- ROC AUC: 0.86

3. Feature Importance

The feature importance was assessed using permutation importance. The plot below shows the importance of each feature in predicting the income level.

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4. Clustering Analysis

KMeans clustering was applied to the dataset to uncover latent patterns. The dataset was reduced to two dimensions using PCA for visualization. Clusters were identified and analyzed to understand their characteristics.